Yujie Xing

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EDUCATIONAL BACKGROUND

University College London (UCL) **09/2026-Present**

- *MSc Artificial Intelligence for Sustainable Development (In progress)*

University of Reading & NUIST **09/2022-06/2026**

- *Bachelor of Science in* ***Data Science*****:** GPA: 79.4/100 (First-class Honnours)
- Modules: Programming in C/C++, Mathematics and Computation, Software Systems Design, Programming in Python, Object-Oriented Programming, Data Structures and Algorithms, Databases.

INTERNSHIP

Intern Engineer, **Kehua Holdings Co., Ltd** **07/2025-08/2025**

- Participated in data cleaning, preprocessing, assisting in modeling for business scenarios, as well as database maintenance and optimization tasks.
- Participated in a production data quality optimization project: experimenting with different imputation methods and feature selection strategies, adopting the sliding window technique and combing with statistical filling to improve data quality.

AWARDS AND HONORS

- Second Prize, Lan Qiao Cup National Competition for Software and Information Technology Professionals (Python, Group B), 05/2025
- Outstanding Student, Nanjing University of Information Science and Technology, 06/2024
- Second-class Scholarship, Nanjing University of Information Science and Technology, 06/2024
- Third-class Scholarship, Nanjing University of Information Science and Technology, 06/2023

RESEARCH EXPERIENCE

Patent: ***A Method for Power Load Forecasting based on the SLSTM-RMTN Model*** **07/2025**

- Utilized STL decomposition of data to separately address trend, seasonal, and residual components.
- Selected target sensors for improved data processing and prediction accuracy.
- Designed and trained a neural network, combining Long Short-Term Memory (LSTM), residual connections, and polynomial expansions for each component.
- Built separate power load forecasting models for each data component.
- Integrated component-specific forecasts to complete the power load prediction for the target power system.
- Achieved accurate capture of time-series characteristics in power load data, reducing data dimensionality.

Development of Intelligent and Rapid Spectrum Resource Scheduling Software **09/2024-06/2025**

- Responsible for the design and training of the core model: building and implementing a Deep Q Network (DQN) model using Python and the PyTorch/TensorFlow framework to address the problem of intelligent spectrum scheduling in complex electromagnetic environments. My main responsibilities included:
- Model design and implementation: defined the state space, action space, and reward function.
- Data preparation and training: utilized the Pandas library to process and integrate historical big data and simulation data for spectrum management, and constructed an experience replay buffer for model training.
- Network training and tuning: performed mini-batch data sampling from the experience replay buffer to update network parameters and used visualization libraries such as Matplotlib to monitor and analyze the training process (e.g., loss function changes, reward curves) in real-time.

Intelligent Prediction of Photovoltaic Power Generation Based on Mechanistic Models and Field Measurements **04/2024 - 04/2025**

- Developed the prediction model using PyTorch/TensorFlow based on the Transformer architecture.
- Implemented a custom self-attention mechanism and an energy conservation constraint layer to effectively enhance the model's efficiency in extracting meteorological features and improve the physical reasonableness of the prediction results.
- Used Pandas and NumPy to integrate and process vast amounts of historical meteorological data and field measurement data from photovoltaic power stations, independently completing tasks including data normalization, time-series missing value interpolation, and the construction of feature-label pairs.
- Adjusted key hyperparameters, including learning rate, network depth, and L2/Dropout regularization.
- **Patent:** ***An Intelligent Prediction Method and System for Photovoltaic Power Generation based on the Integration of Mechanistic Models and Field Measurements***

TECHNICAL SKILLS

- **Programming:** Python, C++, C, Java, SQL, Shell scripting
- **Machine Learning & Deep Learning:** PyTorch / TensorFlow; Transformers, LSTM, DQN
- **Data Analysis & Visualisation:** Pandas, NumPy, Matplotlib, Seaborn